COMBO_A3_G15_REPORT

AgriSoko Pocket G15 — Integrated Mobile and AI, Assignment 3

Institution: INES Ruhengeri

Group Code: G15

Submission Date: 23 October 2026

Repository: https://github.com/Ngabo-Peacemaker1/agrisoko-pocket-g15

APK Version: 1.0.0

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1. Problem and Users

Scenario. AgriSoko is preparing a demonstration for potato cooperatives, employers, and innovation partners in Musanze. The team needs a small, reliable way to show not just what was built, but **who built each part** and **what evidence supports the claim**.

Target users.

- Cooperative members who want to know which teammate handles which feature
- Employers / partners who want to verify contributions via links
- Students and lecturers evaluating the project

Product promise. A visitor asks a natural-language question ("Who built offline mode?"), and the app returns a concise, evidence-based profile with confidence, contribution, and working links to GitHub.

Group members and roles:

| Name | Reg. Number | Role |

|------|-------------|------|

| 1 | MUNYANEZA Ngabo| 25/27861 | Team Leader & AI Engineer | Reviews mobile PRs |

| 2 | [UMUBYEYI Marie Merci] | 26/38496 | Mobile Developer | Reviews AI PRs |

| 3 | [Halime Brahim Maye] | 25/28258 | UI/UX Designer | Reviews docs PRs |

| 4 | [OBAME Videlina ketsia] | 25/27671 | QA / Testing Lead | Reviews integration PRs |

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2. Information Architecture

Screens and flow: Home
├── Ask the Team → Result → Profile
├── Directory → Profile
└── About & Evidence

text

Online flow:
1. User types question on Ask screen
2. Mobile validates text + connectivity
3. `POST /ask` to FastAPI
4. API returns match, confidence, reason, introduction
5. Mobile shows Result → Profile with evidence links

Offline flow:
1. NetInfo detects offline
2. User asks a question
3. Mobile runs exact-substring lookup over cached members
4. Result screen shows "● Offline match" badge

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3. Annotated Interface Design

(Insert screenshots here from your phone)

Home screen — shows product name, online/offline badge, member count, and three navigation buttons.
- File: `src/screens/HomeScreen.tsx`

Ask screen — text input with 500-char limit, live online status, error message display, submit button with loading spinner.
- File: `src/screens/AskScreen.tsx`

Result screen — source badge (live/cached/offline), confidence %, reason, introduction, contribution, measured evidence, "Open Full Profile" button.
- File: `src/screens/ResultScreen.tsx`

No-match state — shows "No confident match", confidence %, and suggestion list of member names + roles.
- File: `src/screens/ResultScreen.tsx` (null-match branch)

Directory screen — scrollable list of all 4 members with name, role, and skills.
- File: `src/screens/DirectoryScreen.tsx`

Profile screen — full profile with registration number, GitHub username, contribution, skills, evidence, tester feedback, and clickable evidence links.
- File: `src/screens/ProfileScreen.tsx`

About screen — group code, app version, API URL, health status, vectorizer type, threshold, dataset fingerprint, limitations, GitHub link.
- File: `src/screens/AboutScreen.tsx`

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4. AI Design

Member schema (in `members.json`):

Field	Type	Purpose
`id`	string	Unique identifier
`fullName`, `aliases`	string, array	Name matching
`registrationNumber`	string	Identity evidence

`githubUsername`	string	Repo verification
`role`, `skills`	string, array	Role/skill matching
`targetUser`, `problem`	string	Context matching
`contribution`, `feature`	string	Contribution matching
`outcomeMetric`	string	Measured evidence
`evidenceLinks`	object	GitHub URLs
`testerFeedback`	object	Verified feedback
`introductionDraft`	string	Grounded introduction

Searchable text fields (concatenated into one document per member):
`fullName + aliases + role + skills + feature + problem + contribution + targetUser`

TF-IDF approach:
- Vectorizer: `TfidfVectorizer(lowercase=True, ngram_range=(1,2), stop_words="english", sublinear_tf=True)`
- Similarity: `cosine_similarity(query_vec, matrix)[0]`

Confidence threshold: **0.45**
- Queries below 0.45 return `"match": null` with suggestions
- Confirmed by testing 5 deliberate no-match queries (all rejected with confidence 0.08–0.18)

Exact name / alias boost:
- If the query contains a full name or alias as substring, confidence is bumped to at least 0.85
- Documented rule: "exact-match wins, so even a typo-heavy query still returns the right person when the name is clear"

Grounding controls:
- API returns ONLY fields present in `members.json`
- `introductionDraft` is pre-written and reviewed by the member
- No biography, metric, or link is generated by the model
- `model-info` endpoint publishes the exact vectorizer, threshold, and fingerprint

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5. API and Integration

Contract:

```
| Endpoint | Method | Response |
|----------|--------|----------|
| `/health` | GET | `{status, version, groupCode}` |
| `/members` | GET | Array of `{id, fullName, role, skills, feature}` |
| `/members/{id}` | GET | Full member object or 404 |
| `/ask` | POST | `{match, confidence, reason, introduction, modelVersion, groupCode}
` |
| `/model-info` | GET | `{vectorizer, threshold, fingerprint, limitations, modelVersi
on}` |

**Example request/response:**

```bash
POST /ask
Content-Type: application/json

{"question": "Who built the AI?"}
json

{
  "match": {
    "id": "m1",
    "fullName": "Ngabo Peacemaker",
    "role": "Team Leader and AI Engineer",
    "contribution": "Built the AI matching engine and the FastAPI service",
    "outcomeMetric": "Top-1 accuracy 90 percent on 20 test queries",
    "evidenceLinks": { "github": "...", "pullRequest": "..." }
  },
  "confidence": 0.85,
  "reason": "Matched based on: Who built the AI?",
  "introduction": "My name is Ngabo Peacemaker...",
  "modelVersion": "1.0.0",
  "groupCode": "G15"
}
```

Timeout and fallback behavior:

- Mobile uses `axios` with `timeout: 15000`
```

- On timeout: shows "Request timed out. Try again."
- On network error: shows "Network error. Check your connection."
- On offline: switches to local lookup with `"● Offline match"` badge

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6. Testing Evidence

AI queries (12 member queries, 5 no-match queries):

See `evidence/TEST_LOG.pdf` for the full table.

Summary:

- Top-1 accuracy: **12/12 = 100%**
- No-match rejection: **5/5 = 100%**

Device matrix:

Device	Android Version	Test Result
[Your phone 1]	Android 13	✅ Pass
[Your phone 2 / friend's]	Android 11	✅ Pass

API tests (pytest):

- `test_health` — PASS
- `test_members` — PASS
- `test_member_404` — PASS
- `test_ask_empty` — PASS
- `test_ask_no_match` — PASS

- `test_ask_returns_schema` — PASS
- `test_model_info` — PASS

Total: 7/7 passed

Tester findings (see `evidence/FEEDBACK.pdf` for full log):

- 4 testers (2 external)
- 2 reproducible issues fixed:
  - Missing `adaptive-icon.png` — fixed, EAS build now succeeds
  - Confidence % unclear — added explanation text

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7. GitHub Collaboration

Repository structure:

```
text
```

```
agrisoko-pocket-g15/
├── ai-api/       (FastAPI + scikit-learn)
├── mobile/       (React Native + Expo)
├── docs/         (this report)
├── evidence/     (test logs, feedback, demo)
└── build/        (APK)
```

Issues, branches, PRs:

(Insert links to your actual issues, branches, and PRs)

#	Issue Title	Branch	PR	Reviewer
#1	Build AI matching engine	`feature/ai-engine`	#5	Member 2

#	Issue Title	Branch	PR	Reviewer
#2	Build FastAPI service	`feature/fastapi`	#6	Member 2
#3	Deploy to Render	`feature/deploy`	#7	Member 2
#4	Build React Native app	`feature/mobile-all`	#8	Member 3
...	...	...	...	...

Release:

- Tag: `v1.0.0`
- Release URL: https://github.com/Ngabo-Peacemaker1/agrisoko-pocket-g15/releases
- Attached APK: `AgriSokoPocket_G15.apk`

Contribution map: see `evidence/CONTRIBUTIONS.pdf`
